Day 8 — When To Use It: Nikhilam vs. Long Multiplication

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will articulate the regime where Nikhilam is faster than long multiplication, AND will recognise problems where long multiplication (or *Ūrdhva-tiryagbhyāṁ*, the general "vertically and crosswise" method) is the better choice.

Materials

- Whiteboard, two columns labelled "Use Nikhilam" and "Use long / Ūrdhva."
- "Decide-or-don't" card deck — 20 multiplication problems on cards. (Teacher prepares ahead.)
- Reflection slip.

Lesson flow

Warm-up (5 min)

- Quick Nikhilam: `97 × 96` (9312), `104 × 103` (10712), `102 × 98` (9996).
- *"Now — would Nikhilam work for `47 × 53`? Let's see."*

Core (15 min) — When it works, when it doesn't

1. **The honest claim.** Nikhilam multiplication is fast *only* when both factors are close to the same power of 10.

2. **Try a "bad" example: `47 × 53`:**

    - Base = ? Neither factor is near 10 (too big), near 100 (too far below), or near 50 (a *sub-base*; advanced).
    - If you force base 50: deficits +(−3), +3. Left = 47 + 3 = 50. Right = (−3)(+3) = −9. Answer = 50 × 50 − 9 = 2491. ✓ Slow because you have to compute 50 × 50 = 2500 = a *multiplication*.
    - **Sub-bases work** (Day 8 tier-up), but they're not faster than long multiplication for casual users.
3. **A genuinely bad case: `47 × 89`:**

    - Not near any common base.
    - Use long multiplication. **It's faster.**
4. **The decision rule (write on board, boxed):**

    Use Nikhilam when:
    - Both factors are within ~5 of a power of 10 (10, 100, 1000): trivially fast.

    - Both within ~10–15 of a power of 10: still faster than long multiplication for most students.

    Use long multiplication (or *Ūrdhva*) when:

    - One factor is “random” (not near a base).
    - The factors are near different bases.
    - Speed doesn’t matter (writing out long multiplication is more error-resistant).

5. **The contrast with Module 7 (×11) and Module 9 (subtraction).**

    - Module 7’s ×11 trick works *always* when one factor is 11. It’s a different niche.
    - Module 9’s subtraction-by-complement (Nikhilam again) works when subtracting from a power of 10.
    - Nikhilam multiplication is just one shortcut in a toolkit. Long multiplication remains the workhorse.

6. **The other Tirthaji sūtra mentioned today: *Ūrdhva-tiryagbhyāṁ*** (“vertically and crosswise”) is Tirthaji’s sūtra #3 — the **general** multiplication procedure. It works for any pair of numbers. We won’t drill it in this module; it’s a topic of its own. Just notice: *Ūrdhva* covers what Nikhilam can’t.

Activity (20 min) — “Decide-or-don’t” game

Rules: - Pairs receive a deck of 20 cards. Each card has one multiplication problem. - For each card, the pair must **first decide** in 5 seconds: Nikhilam or long? - Then they solve it. - If they used Nikhilam and the method actually applies: full marks. - If they used long multiplication and Nikhilam would have been visibly faster: half marks. - If they used Nikhilam where it wasn’t suitable and got stuck: zero marks (but learning is the lesson).

Sample cards:

```
97 × 96       → Nikhilam (both near 100)
47 × 53       → Long (no common base)
998 × 997     → Nikhilam (near 1000)
73 × 84       → Long
103 × 104     → Nikhilam
89 × 91       → Long-ish (near 90, but no nice base)
102 × 99      → Nikhilam (mixed)
347 × 28      → Long
988 × 994     → Nikhilam (near 1000, larger deficits but still works)
56 × 89       → Long
99 × 99       → Nikhilam (101 × 99 would also be nice)
1003 × 1004   → Nikhilam
65 × 75       → Long
107 × 108     → Nikhilam
1234 × 9876   → Long (too far apart, too random)
9999 × 9998   → Nikhilam (near 10000)
112 × 113     → Nikhilam (with carry)
156 × 178     → Long
49 × 51       → Nikhilam (both near 50 if comfortable; else long)
997 × 1003    → Nikhilam (mixed, near 1000)
```

- 20 cards in 20 minutes. Pairs that finish early swap decks with another pair.

Wrap-up (5 min)

- Class discussion: which cards did pairs most often mis-categorise?
- Insight to leave with: **the method is a tool, not a rule.** Long multiplication is *not* “the wrong way.” Each tool fits a certain shape of problem.
- Preview Day 9: speed sprints + project start.

Student-facing content

When to use Nikhilam — and when not to

Use Nikhilam	Use long multiplication
Both factors within ~10 of 10, 100, 1000, 10000	One factor “random” (not near a base)
Mental math, no pencil	Pencil work, no time pressure
Mixed below/above same base	Factors near *different* bases

The honest truth: Most multiplication problems in life are NOT Nikhilam-shaped. Long multiplication remains your everyday workhorse.

But when both factors *are* near a base — `97 × 96`, `1003 × 1004` — Nikhilam is 5–10× faster.

Tirthaji’s other multiplication sūtra, *Ūrdhva-tiryagbhyāṁ* (“vertically and crosswise”), works for any pair of numbers and is taught in its own module. Nikhilam is the *near-base specialist*; *Ūrdhva* is the *generalist*.

Homework

- Find 5 multiplication problems in your school textbook (any subject). For each, decide: Nikhilam, long, or Ūrdhva. Write 1 sentence justifying your choice. Bring tomorrow.

Differentiation

- **Tier up:** Try a sub-base (base 50). Compute `47 × 48` via base 50: deficits 3, 2. Cross: $47 - 2 = 45$. Multiply by 50: 2250. Add deficit product 6 → 2256. (Real: $47 \times 48 = 2256$. ✓) Compare with long multiplication. Which felt faster?
- **Tier down:** Stick to clear-cut cases. Distinguish: “is this near 100?” If yes, Nikhilam. If no, long.

Teacher notes

- This is the honesty day. Some students will be disappointed that Nikhilam isn’t universally faster. That’s the point: claims about Vedic mathematics often overstate the universal advantage. The honest framing protects against later disillusionment.
- The “decide” step before solving is the habit-of-mind worth building. Without it, students apply Nikhilam to inappropriate problems and get slower than the standard algorithm.
- The mention of *Ūrdhva-tiryagbhyāṁ* sets up future modules. Don’t elaborate on the procedure today — it’s a 2-week topic of its own.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam scope and limits.
- *Microsoft Word - sutras1.pdf* — sūtras #2 (Nikhilam) and #3 (Ūrdhva-tiryagbhyāṁ).